

MCQ

Topic: Data Types (List, Dictionary, Tuple, Set), Operators & Precedence

Student Name: _____ Date: _____ Score: _____

Instructions: Select the correct option by filling or marking the checkbox ☐ next to it.

1. Which of the following statements is true about Python tuples?

- ☐ Tuples are mutable, meaning their size and values can be modified after creation.
- ☐ A tuple can contain mutable objects, such as lists, as its elements.
- ☐ Tuples can only store data elements that share the exact same data type.
- ☐ An empty tuple is initialized using square brackets [].

2. What is the evaluated output of the following mathematical expression in Python?

```
result = 5 + 2 ** 2 * 3 / 4
```

- ☐ 2.0
- ☐ 13.5
- ☐ 8.0
- ☐ 10.0

3. What is the 8-bit two's complement representation of the decimal number -45?

- ☐ 00101101
- ☐ 11010010
- ☐ 11010011
- ☐ 11101101

4. What will be displayed on the console after running the following script?

```
list_a = [1, 2, 3]
list_b = list_a
list_b.append(4)
print(list_a)
```

- ☐ [1, 2, 3]
- ☐ [1, 2, 3, 4]
- ☐ [1, 2, 3, [4]]
- ☐ [4, 1, 2, 3]

5. Which syntax returns elements that exist in `set_a` but are completely absent from `set_b`?

- ☐ `set_a - set_b`
- ☐ `set_a & set_b`
- ☐ `set_a | set_b`
- ☐ `set_a ^ set_b`

6. What is the evaluation outcome of the list expression: `[1, 2] * 2 + [3]`?

- ☐ [1, 2, 1, 2, 3, 3]
- ☐ [1, 2, 1, 2, 3]
- ☐ [[1, 2], [1, 2], 3]
- ☐ [2, 4, 3]

7. Given `my_dict = {'a': 1, 'b': 2}`, what does `my_dict.get('c', 0)` return?

- ☐ None
- ☐ Raises a `KeyError`
- ☐ 0
- ☐ 'c'

8. What will be output of

```
s = {True, False, 0, 1, 2, 3, 3}
print(s)
```

- ☐ {0,1,2,3}
- ☐ {True,False,2,3}
- ☐ {0,1,True,False,2,3}
- ☐ {True,False,0,1,2,3,3}

9. What is the final evaluation value of `result` below?

```
x = 0
print((x > 0) and (10 / x > 1))
```

- ☐ True
- ☐ False
- ☐ Raises a `ZeroDivisionError`
- ☐ 0

10. Which is the Output?

```
s = [1,2,3]
a= [1,2,3]
print((s is a) and (s ==a) or ('1' in a))
```

- ☐ True
- ☐ False

11. What is the minimum memory size in bytes occupied by a basic integer object and Integer Object?

- ☐ 24 bytes and 28 bytes
- ☐ 28 bytes and 24 bytes
- ☐ Both 24 bytes
- ☐ Both 28 bytes

12. Which operator possesses the highest precedence among the following choices?

- ☐ + (Addition)
- ☐ ** (Exponentiation)
- ☐ * (Multiplication)
- ☐ not (Logical NOT)

13. What is the resulting configuration of the list after this operation?

```
nums = [1, 2, 3]
nums.extend([4, 5])
print(nums)
```

- ☐ [1, 2, 3, [4, 5]]
- ☐ [1, 2, 3, 4, 5]
- ☐ [5, 4, 1, 2, 3]
- ☐ [9, 3]

14. What happens when evaluating the mathematical modulo expression `-5 % 2` in Python?

- ☐ -1
- ☐ 1
- ☐ 2.5
- ☐ Raises a `ArithmeticError`

15. What is the length of the set generated by this code?

```
sample_set = {1, 2, 2, '2', (1, 2)}
print(len(sample_set))
```

- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 2

16. What value is bound to `ans` after assessing the order of operations below?

```
ans = 2 ** 3 ** 2 - 4 * 3 // 2 > 100
```

- ☐ 506
- ☐ True
- ☐ False
- ☐ 58

17. Which statement correctly throws an error when attempted on a dictionary?

```
d = {'x': 10}
```

- ☐ `d['y'] = 20`
- ☐ `print(d['y'])`
- ☐ `d.clear()`

☐ `del d['x']`

18. What will be printed by the comparison chaining expression `1 < 3 > 2`?

☐ `True`

☐ `False`

☐ Raises a `SyntaxError`

☐ `None`

19. How do you successfully remove all structural elements from a mutable Python set named `s`?

☐ `s.remove()`

☐ `s.clear()`

☐ `s.delete()`

☐ `s = []`

20. What is the computed result of the expression `-12`?

☐ `11`

☐ `-11`

☐ `10`

☐ `-10`